E-classroom	**Name:**					
	Gr 10		**Date:**		**Time:**	$2\frac{1}{2}$ **HR**
CAPS Reference	**Function**					
Topic	**Exponential Function**					

1. GOT IT! [15 MIN]

Basic standard:

$$f(x) = ab^x + q$$

$f(x)$: – represents the y – value

a: – indicates gradient and if the function is positive or negative

b: – base of the exponent

$y = 0$ – indicates the $x - intercept$

$x = 0$ – indicates the $y - intercept$

q: – indicates the horizontal asymptote $y = q$

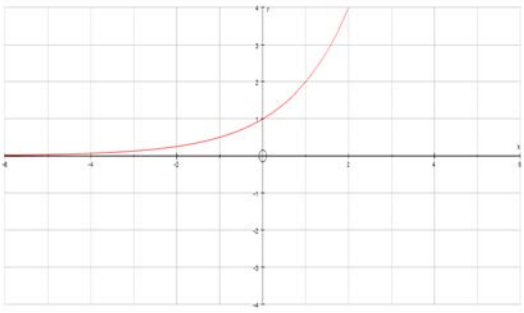
$\therefore q$ – indicates when the function will move up or down

Exponential function.

Positive function; $a > 0$

Domain: $x \in (-\infty; \infty)$

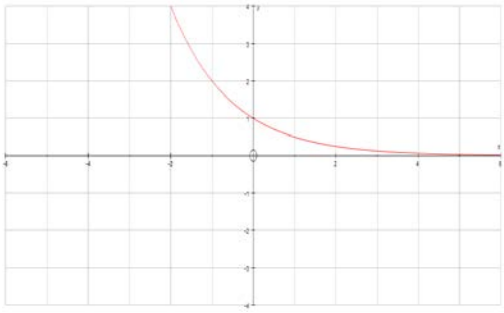
Range: $y \in (q; \infty)$



Negative function; $a < 0$

Domain: $x \in (-\infty; \infty)$

Range: $y \in (q; \infty)$



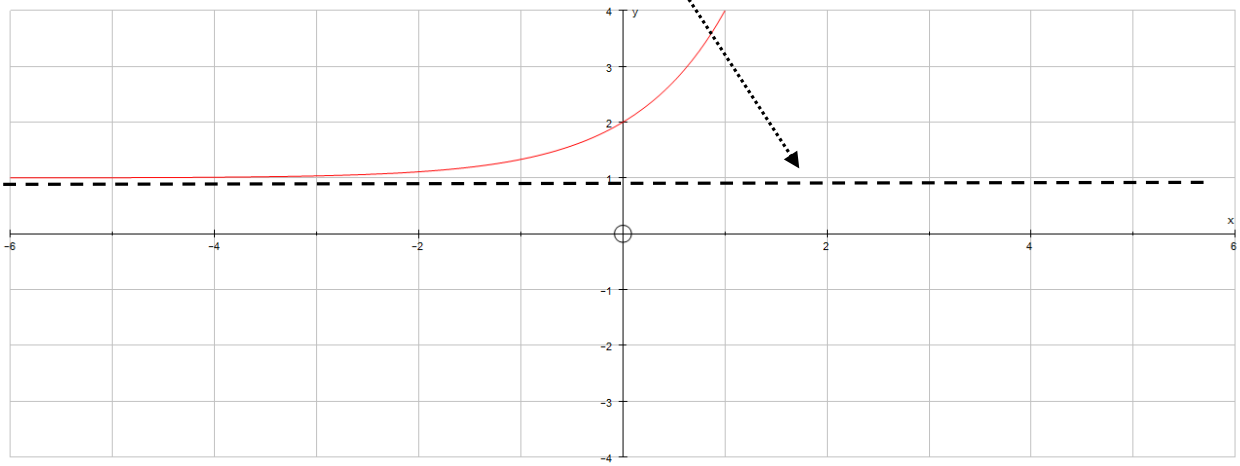
Drawing of an Exponential function.

$$f(x) = 3^x + 1$$

Most important!!

Asymptote

$y = 1$



For the Exponential function make use of a table

x	-2	-1	0	1	2
$3^x + 1$	$\frac{10}{9}$	$\frac{4}{3}$	2	4	10

Important information about the function.

$a > 0$

This indicates that the function is positive.

$q = 1$

This indicates that the function has moved 1 units up from the original function $y = b^x$

Asymptote $y = 1$. There for the function does not touch the line $y = 1$

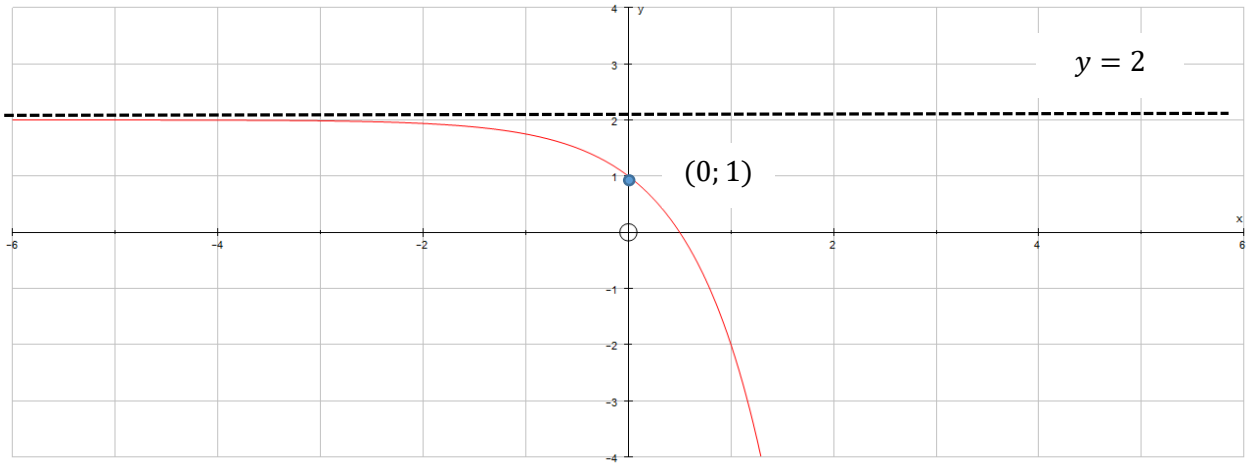
Domain: $x \in (-\infty; \infty)$

Range: $y \in (1; \infty)$

Determine the equation of an Exponential function

$$\boldsymbol{f(x) = ab^x + q}$$

Given: $\boldsymbol{f(x) = a.4^x + q}$



Step 1: Substitute the asymptotes $y = 2$

$y = a.4^x + 2$

Step 2: Substitute the given coordinate $(0; 1)$

$y = a.4^x + 2$

$1 = a.4^0 + 2$

$1 - 2 = a$

$-1 = a$

Step 3: Substitute a into the equation.

$y = -4^x + 2$

$$\boldsymbol{f(x) = -4^x + 2}$$

2. GO AHEAD [60 MIN]

2. Sketch the following on a set of axis and answer the questions that follows.

2.1 $f(x) = 2^x$

a) Write down the range of f

b) Write down the asymptote of f

c) Determine the value of x for which $f(x) > 0$

d) Write down the equation of h, where $h(x) = f(x) - 1$

2.2 $g(x) = \left(\frac{1}{2}\right)^x$

a) Write down the range of f

b) Write down the asymptote of g

c) Determine the value of x for which $g(x) > 0$

d) Write down the equation of h, where $h(x) = g(x) + 1$

2.3 $f(x) = 3^x + 2$

a) Write down the range of f

b) Write down the asymptote of f

c) Determine the value of x for which $f(x) > 0$

d) Write down the equation of h, where $h(x) = -f(x)$

2.4 $g(x) = 4^x - 1$

a) Write down the range of g

b) Write down the asymptote of g

c) Determine the value of x for which $g(x) > 0$

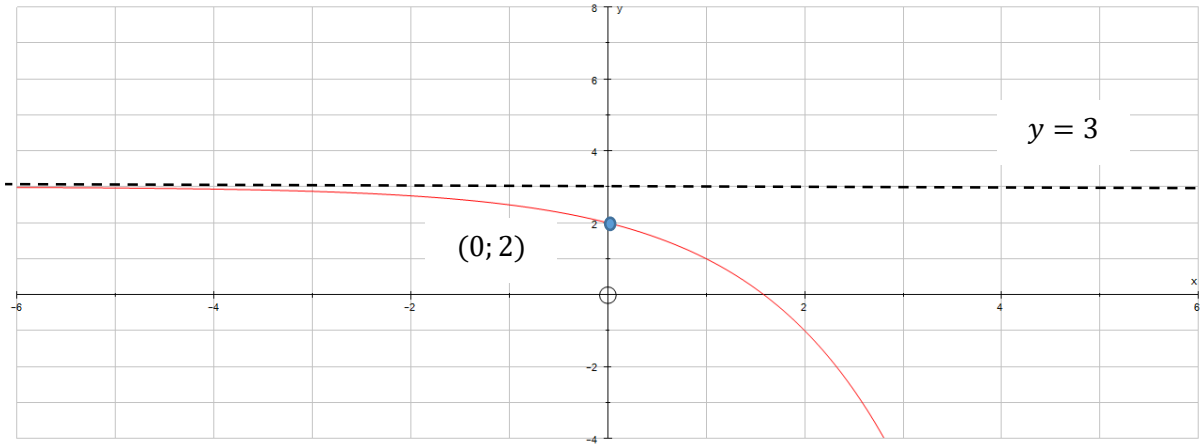
d) Write down the equation of h, where $h(x) = g(x) - 2$

3. CHECK YOUR WORK![15 MIN]

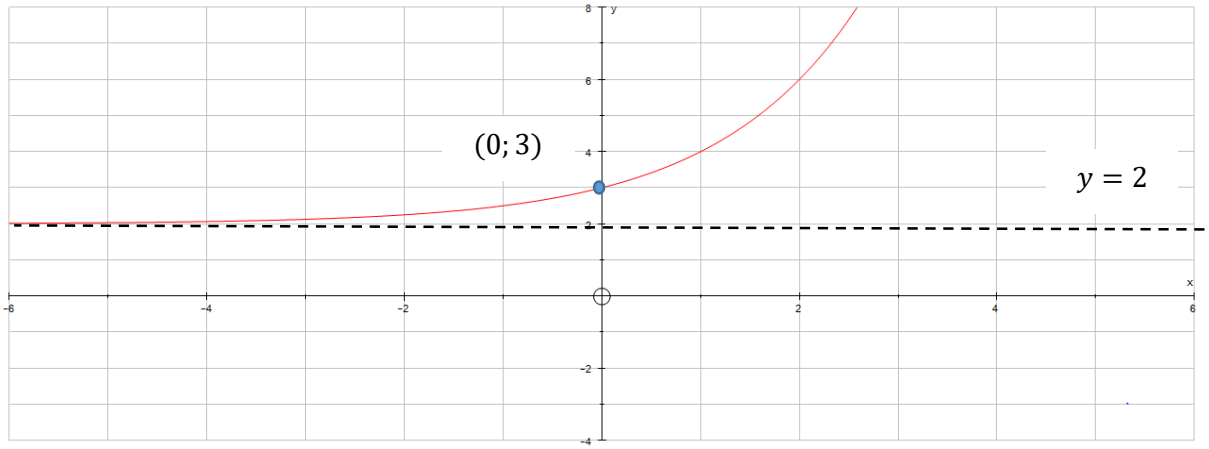
4. GO AHEAD [60 MIN]

4. Determine the equation of the following functions:

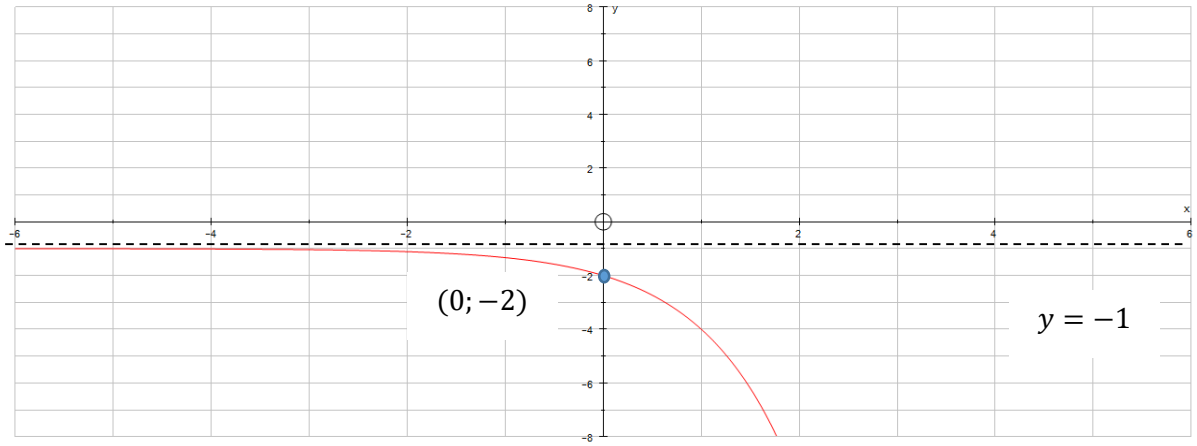
4.1 $f(x) = a.2^x + q$



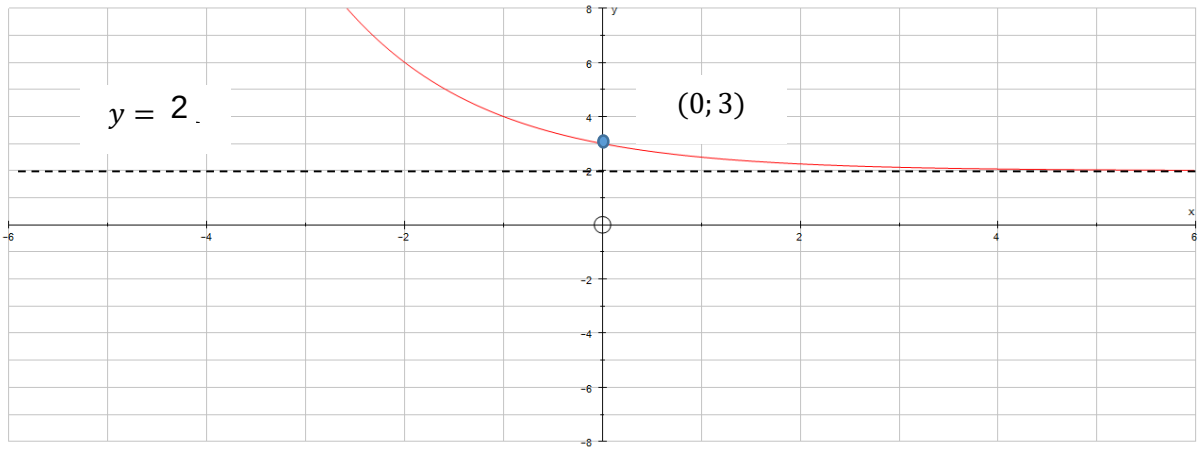
4.2 $f(x) = a.2^x + q$



4.3 $f(x) = a.3^x + q$



4.4 $f(x) = a\left(\frac{1}{2}\right)^x + q$



5. CHECK YOUR WORK! [15 MIN]